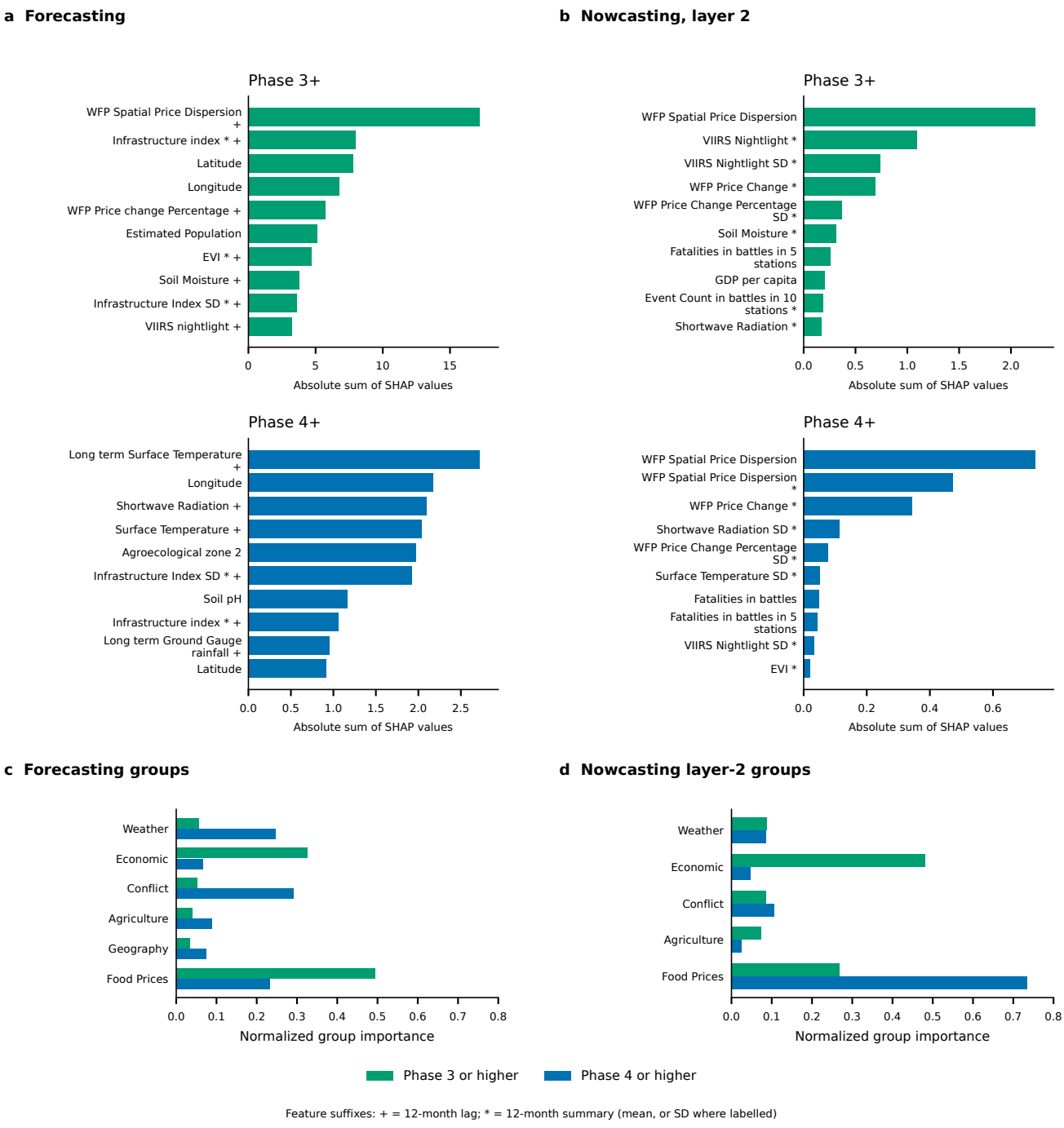
a Forecasting
Phase 3+
WFP Spatial Price Dispersion +
Infrastructure index * +
Latitude
Longitude
WFP Price change Percentage +
Estimated Population
EVI * +
Soil Moisture +
Infrastructure Index SD * +
VIIRS nightlight +
0
5
10
15
Absolute sum of SHAP values
Phase 4+
Long term Surface Temperature +
Longitude
Shortwave Radiation +
Surface Temperature +
Agroecological zone 2
Infrastructure Index SD * +
Soil pH
Infrastructure index * +
Long term Ground Gauge rainfall +
Latitude
0.0
0.5
1.0
1.5
2.0
2.5
Absolute sum of SHAP values
b Nowcasting, layer 2
Phase 3+
WFP Spatial Price Dispersion
VIIRS Nightlight *
VIIRS Nightlight SD *
WFP Price Change *
WFP Price Change Percentage SD *
Soil Moisture *
Fatalities in battles in 5 stations
GDP per capita
Event Count in battles in 10 stations *
Shortwave Radiation *
0.0
0.5
1.0
1.5
2.0
Absolute sum of SHAP values
Phase 4+
WFP Spatial Price Dispersion
WFP Spatial Price Dispersion *
WFP Price Change *
Shortwave Radiation SD *
WFP Price Change Percentage SD *
Surface Temperature SD *
Fatalities in battles
Fatalities in battles in 5 stations
VIIRS Nightlight SD *
EVI *
0.0
0.2
0.4
0.6
Absolute sum of SHAP values
c Forecasting groups
Weather
Economic
Conflict
Agriculture
Geography
Food Prices
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Normalized group importance
d Nowcasting layer-2 groups
Weather
Economic
Conflict
Agriculture
Food Prices
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Normalized group importance
Phase 3 or higher
Phase 4 or higher
Feature suffixes: + = 12-month lag; * = 12-month summary (mean, or SD where labelled)